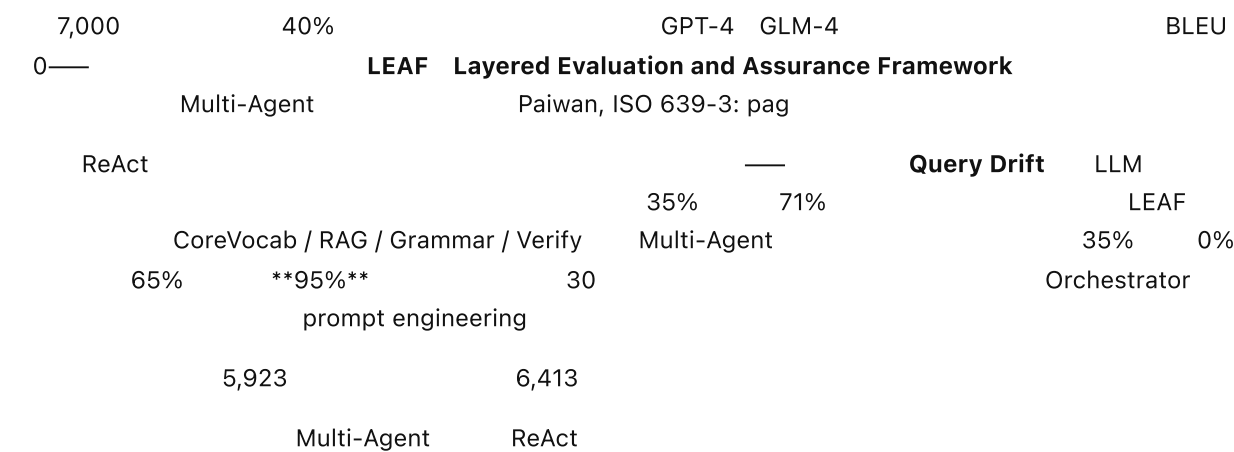




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1

1.1



1

			BLEU	
GLM-4-Flash	→	40	0.000	0.0%
GLM-4-Plus	→	40	0.000	2.5%
ICL 5-shot	→	40	0.000	0.0%
ICL 10-shot	→	40	0.000	0.0%
ICL 20-shot	→	40	0.000	7.5%

20 ICL BLEU 0 LLM In-Context Learning

1.2 RAG

RAG 242 RAG 90% 2

2 RAG

		BLEU	
RAG	242	0.452	90.0%
RAG	242	0.416	85.0%

RAG	4,844	0.873	77.5%
-----	-------	-------	-------

RAG

" "→"

"

—

LLM

ReAct

1.3

1.		ReAct	Agent	LLM	35%	71%
2.	LEAF		+ Multi-Agent			0%
3.	8	Truth Table +				

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2

	NLP				Koehn et al., 2007	
					Conneau et al., 2020	
RAG			Lewis et al., 2020		LLM	RAG
			" "		LLM	
Multi-Agent	AutoGen	Wu et al., 2023	CrewAI	Agent		
	Agent		Multi-Agent			
ReAct	Yao et al. (2023)	ReAct	Reasoning + Acting	LLM		
	ReAct " "					

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3

3.1

LEAF	Multi-Agent	4	Agent		
	• Orchestrator Agent		ReAct	Agent	
	• Knowledge Agent				RAG + CoreVocab + ExactMatch
	• Quality Agent				Reverse Verification
	• Teaching Agent				

Agent

AgentMessage

MessageBus

3.2

L1	CoreVocab		+
L2	RAG		+ FAISS

L3	Grammar		$tj \rightarrow t, lj \rightarrow l \quad +$
L4	Verify		

3.3

ReAct Agent LLM Query

Drift

$$\text{Drift}(q_{\text{orig}}, q_{\text{retrieval}}) = \mathbb{1}[q_{\text{orig}} \neq q_{\text{retrieval}}]$$

q_{orig} $q_{\text{retrieval}}$

	→	LLM
	→ →	LLM
	→ →	LLM

3.4

LEAF prompt engineering

1. Orchestrator system prompt " translate text "

2. Orchestrator schema LLM

3. translate → rag_search → lookup

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4

4.1

- Paiwan, ISO 639-3: pag 5,923
- Amis, ISO 639-3: ami 6,413
- GLM-4-Flash AI
- 20 → benchmark_v2.json
- BLEU Exact Match

4.2 A 8 Truth Table

" " " Multi-Agent" 3

3 8

#			/20	
1	LLM Direct	0%	0/20	—
2	RAG Only	80%	16/20	+80pp

3	Single Agent ReAct	80%	16/20	0pp
4	SA + ReAct free	65%	13/20	-15pp
5	SA + Constrained ReAct	~80%	12/15	+15pp vs #4
6	Multi-Agent w/ SA prompt	90%	18/20	+25pp
7	Multi-Agent w/o Quality	95%	19/20	+5pp
8	Multi-Agent	95%	19/20	0pp

" #3→#4 — ReAct 15pp LLM "

" #4→#6 +25pp Orchestrator

4.3 B

4

	/ 20		→
SA ReAct	0/20	0%	N/A
SA + ReAct free	7/20	35%	71% 5/7
SA + Constrained ReAct	6/20	30%	—
Multi-Agent + ReAct	0/20	0%	N/A

Constrained Prompt " " 30% prompt engineering

4.4 C

5

Multi-Agent

		Δ vs	
—	95%	—	—
Quality Agent	95%	0pp	
Pre-routing	95%	0pp	
System Prompt SA prompt	90%	-5pp	prompt 5pp
Architecture Orchestrator→SA+ReAct	65%	-30pp	

95% +25pp Orchestrator prompt +5pp 0pp

4.5 D

6

		Δ		Δ
Control	79.2%	—	73.5%	—

L1 CoreVocab	45.8%	-33.4pp	38.8%	-34.7pp
L3 Grammar	75.0%	-4.2pp	73.5%	0pp
L4 Verify	83.3%	+4.1pp	73.5%	0pp

L1 CoreVocab33-35ppControl

5.7pp79.2% vs 73.5%

4.6

E

Query Drift

B

6

Query Drift

Prompt-only LLM	0/6
LEAF Orchestrator + ReAct	6/6

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5

5.1

Prompt Engineering

LLM

—

Constrained Prompt

35%

30%

4

ReAct

LLM

"

"

prompt

LEAF Orchestrator

•

Orchestrator

"

"

Knowledge Agent

"

"

Orchestrator

•

Agent

•

5.2

5,923 vs 6,413

CoreVocab

-33pp vs -35pp

5.3

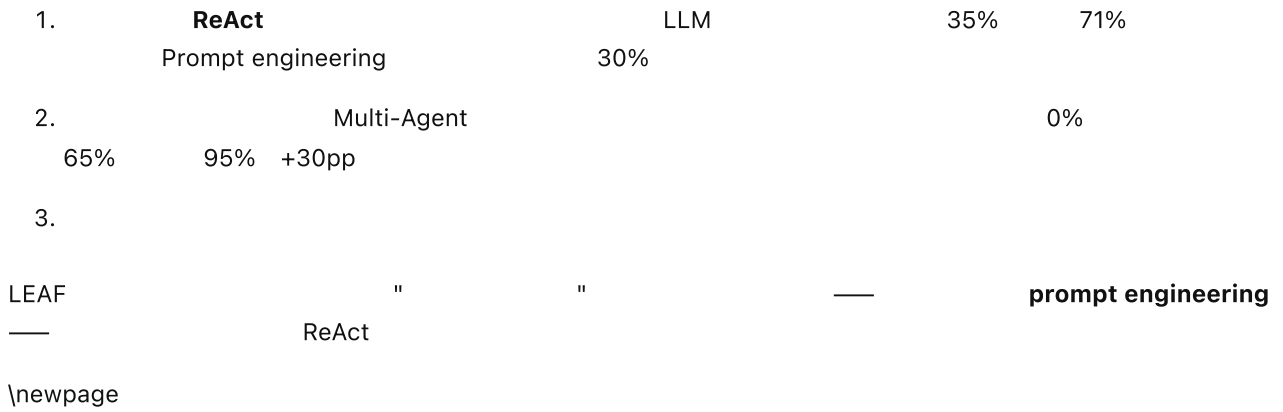
1.
- 20
2.
- GLM-4-Flash
- L4
3.
-

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6

LEAF

ReAct



1. Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., ... & Stoyanov, V. (2020). Unsupervised cross-lingual representation learning at scale. *ACL 2020*.
2. Koehn, P., Hoang, H., Birch, A., Callison-Burch, C., Federico, M., Bertoldi, N., ... & Herbst, E. (2007). Moses: Open source toolkit for statistical machine translation. *ACL 2007*.
3. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., ... & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *NeurIPS 2020*.
4. Wu, Q., Bansal, G., Zhang, J., Wu, Y., Li, B., Zhu, E., ... & Awadallah, A. H. (2023). AutoGen: Enabling next-gen LLM applications via multi-agent conversation. *arXiv preprint arXiv:2308.08155*.
5. Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., & Cao, Y. (2023). ReAct: Synergizing reasoning and acting in language models. *ICLR 2023*.

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A

A.1 Agent

Agent AgentMessage

AgentMessage:

- type: MessageType (TRANSLATE / RETRIEVE / VERIFY / EVALUATE / ERROR)
- sender: str (agent role)
- receiver: str (target agent role)
- content: dict (structured data, NOT natural language)
- metadata: dict (trace info, timestamps)

MessageBus

Agent

A.2 Orchestrator ReAct

1. → q_orig
2. LLM +
 - translate(text=q_orig, direction="c2p") ←
 - rag_search(query=q_orig) ←
 - lookup(word=...) ← RAG
3. →

4.

→
- ReAct
5.

3

A.3 Quality Agent Pipeline

1.

candidate = "kina"

"

"
2.

kina → (

→

) → "

"
3.

≈

"

" → PASS
4.

candidate = "tjina"
- tjina → "???" ≠ "

" → FAIL →

B

B.1 Query Drift

#						SA+ReAct	MA
q06				qiljas	✗	✗	✓
q12				ama	✗	✗	✓
q13				aljak	✗	✗	✓
q16				caucau	✗	✗	✓
q17				kabang	✗	✗	✓
q19				ngadan	✗	✗	✓

6 Multi-Agent

B.2

50 klokah

	39.6/50	79.2%
- CoreVocab	22.9/50	45.8%
- Grammar	37.5/50	75.0%
- Verify	41.7/50	83.3%

	36.8/50	73.5%
- CoreVocab	19.4/50	38.8%

- Grammar	36.8/50	73.5%
- Verify	36.8/50	73.5%

C

GitHub: <https://github.com/yoanahappy2/LEAF>

```
LEAF/
├── src/                                #
│   ├── core/                          # BaseAgent, MessageBus, ReAct Loop
│   ├── agents/                        # Orchestrator, Knowledge, Quality, Teaching
│   └── rag_service.py                 # RAG
├── benchmark/                         #
│   ├── run_baseline.py               #
│   ├── run_rag_benchmark.py          # RAG
│   ├── run_benchmark_v2.1.py         # Multi-Agent
│   ├── metrics.py                   # BLEU + Exact Match
│   └── results/                      # (JSON)
├── experiments/                       #
│   ├── ablation_v4.py               #
│   ├── agent_ablation.py            # Agent
│   ├── constrained_react.py          # ReAct
│   └── critical_experiment.py         # Query Drift
├── data/                              #
│   ├── merged_corpus.json            # (4,844 )
│   └── grammar_rules.json            #
└── docs/                              #
```